DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Anonymous CVPR submission

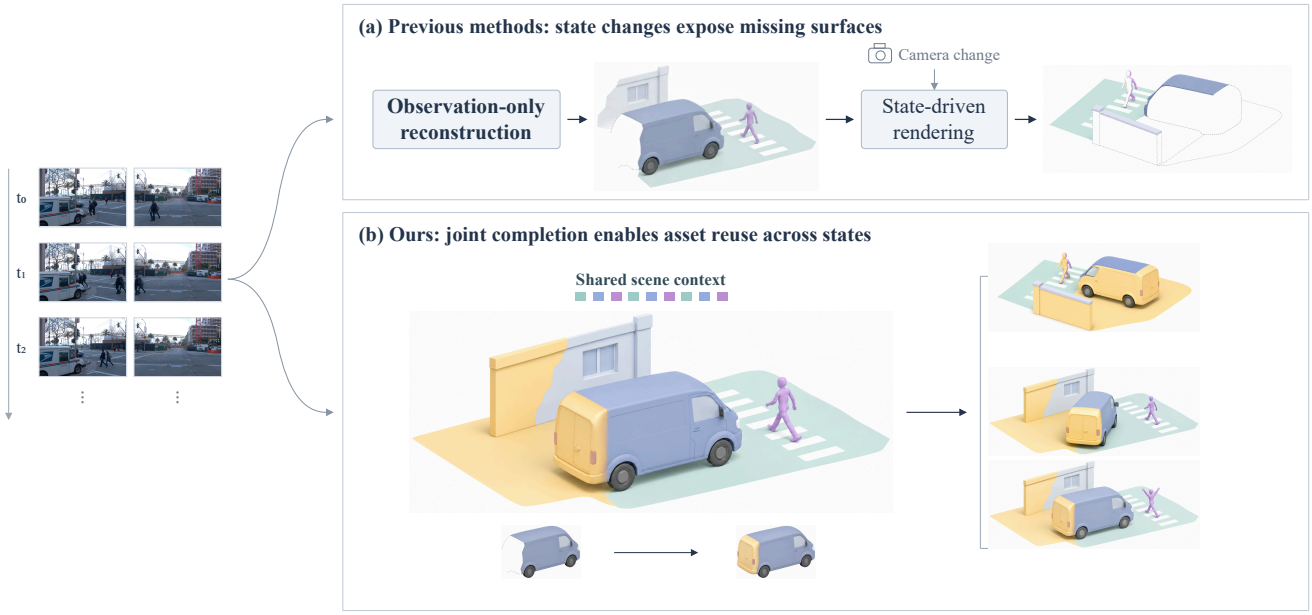

Figure 1. **From observations to reusable completed assets.** Top: viewpoint and actor changes expose missing surfaces in observation-only reconstruction. Bottom: structured encoding and joint completion produce background, canonical vehicles, and humans for reuse across states. Gold indicates completed regions. All scene and asset images are schematic illustrations.

Abstract

Moving the camera or changing actor poses exposes surfaces that were never observed in the input driving images. Reconstructing scenes for simulation and editing therefore requires completing these surfaces while retaining control over actor motion. We present DriveCompose, a framework for compositional generative reconstruction from calibrated driving image sequences. Visual evidence is aggregated in world coordinates for the background and canonical frames for vehicles and humans. A conditional flow model uses cross-component memory to share scene context and jointly complete missing geometry and appearance in the component latents. The completed latents are decoded into Gaussian components with rigid and skeletal bindings, allowing generated surfaces to move with their corresponding actors. The same background and actor assets can then be assembled and rendered under new camera viewpoints and actor states. Experiments on Waymo show higher reconstruction accuracy and better off-trajectory rendering quality than feed-forward reconstruction baselines. Actor editing experiments further demonstrate that completed vehicles and humans can be reused across motion and pose changes.

1. Introduction

Driving simulation and scene editing require 3D scenes that support changes in camera viewpoint, vehicle motion, and human pose [50, 53]. However, driving images reveal only the surfaces visible from the recorded viewpoints and actor configurations [1, 35]. Moving a camera, turning a vehicle, or changing a pedestrian's pose exposes previously unobserved regions (Fig. 1). Reconstructing scenes for these operations therefore requires completing the missing geometry and appearance while ensuring that completed vehicle and

human surfaces move with their respective actors.

Recent feed-forward methods fuse multiple images to directly predict driving-scene geometry and Gaussian representations [5, 8, 49, 52]. Motion and visibility estimates align dynamic objects and aggregate surfaces visible across the input sequence. Surfaces absent from the entire sequence call for generative priors over geometry and appearance.

Generative reconstruction uses asset priors, Gaussian completion, or generated views to infer unobserved content [21, 48, 51, 55]. Compositional reconstruction represents the background and moving actors separately to support vehicle motion and human pose control [7, 47, 53]. Completing these component representations lets generated surfaces retain their actors' motion bindings. The visible background and other actors provide scene context for completing a partially observed component.

We introduce DriveCompose, which jointly completes the background, vehicles, and humans as reusable Gaussian components (Fig. 2). The background is represented in world coordinates, rigid assets in their own frames, and human surfaces in a reference pose with skeletal bindings [7, 47]. Cross-component memory shares these components' features to condition flow-based completion [22]. Gaussian decoders preserve their ownership and bindings [18], allowing completed surfaces to move with their actors under new camera and actor states (Secs. 3.1 and 3.4).

Experiments on Waymo show higher reconstruction accuracy and better off-trajectory rendering quality than the feed-forward baselines. Paired synthetic tests compare joint completion with scene-level generation and cross-object coupling, while real actor edits demonstrate reuse of the same completed assets (Sec. 4).

Our contributions are:

- We introduce DriveCompose to jointly complete backgrounds, vehicles, and humans from image sequences as compositional Gaussian assets, reusable under camera, rigid, and skeletal state changes.
- We develop a cross-component latent completion model based on conditional flow matching. Local attention models spatial relationships within each component, while cross-component memory aggregates background and actor features to provide scene context for generating each component's missing content.
- We establish the contribution of shared scene context through matched completion mechanisms and visibility-stratified evaluations, and demonstrate state-driven reuse with real actor edits.

2. Related work

Driving-scene reconstruction. Urban reconstruction builds on NeRF and Gaussian radiance representations [18, 26], with driving-specific rendering developed in NeuRAD and SplatAD [13, 37]. Feed-forward methods predict scene geometry and Gaussians directly from images [3, 6, 14, 39, 40]. STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with visibility and motion-guided interpolation [5, 49]. Surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction are explored by DrivingForward, StreetForward, PointForward, and Instant NuRec [8, 16, 36, 52]. TokenGS uses learnable tokens to decouple Gaussian prediction from pixels [33].

Compositional scene representations. Separating background and moving components supports actor control in Neural Scene Graphs, Street Gaussians, and DrivingGaussian [27, 47, 54]. OmniRe represents rigid and articulated actors in canonical frames, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [7, 50, 53]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [15, 19, 24, 29, 31]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [25, 32, 56]. These frames and bindings provide the interface for reusing surfaces after reconstruction or completion.

Generative reconstruction and completion. Generative reconstruction infers unobserved geometry and appearance through generated views [11, 23, 34] or structured latent representations [45, 46]. GenAssets learns priors for driving assets, CompleteSplat generates hidden Gaussian surfaces, and JRM couples object latents to aggregate unaligned observations [21, 44, 51]. LatentWorld organizes semantic occupancy generation around entity and background latents [28]. DriveCompose jointly completes world-space background and canonical rigid and human codes, retaining their geometry, appearance, and motion bindings for state-driven rendering.

Driving-scene generation. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [10, 12, 41]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction, and HorizonForge combines scene editing with video diffusion [42, 48]. WorldSplat generates Gaussians and refines their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [43, 55]. FreeGen co-trains feed-forward reconstruction with diffusion-based generation, distilling generative priors into reconstruction for off-trajectory rendering [4]. DriveCompose stores the completed background and canonical assets, then renders the same 3D content under new camera and actor states.

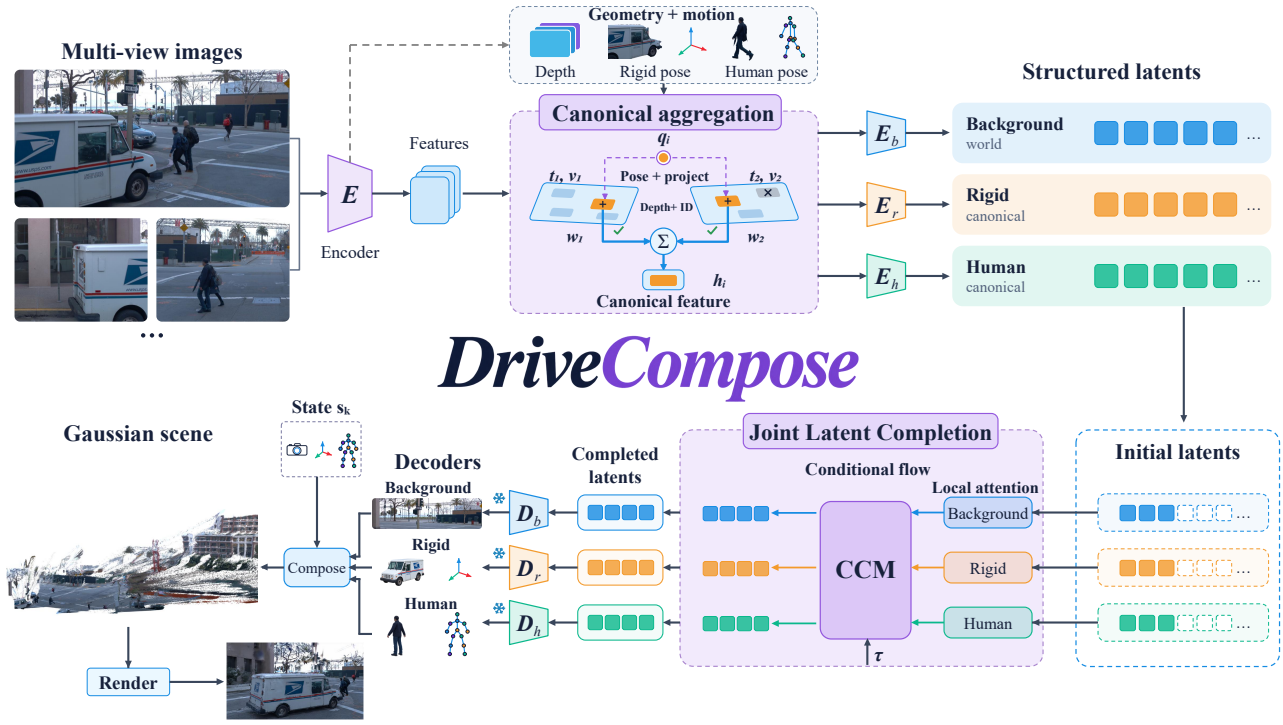

Figure 2. **DriveCompose overview.** Visual encoding and canonical aggregation recover component features; type-specific encoders, joint completion, and Gaussian decoders produce background, vehicle, and human content for state-driven composition. Scene and asset images are schematic illustrations.

3. Method

DriveCompose aggregates observations in world and canonical coordinates, jointly completes their latents, and decodes reusable Gaussian assets for rendering under new states (Fig. 2).

3.1. Problem formulation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. The controlled evaluations provide rig poses, instance identities, vehicle boxes and root trajectories, and target human joints. Human shape and binding are recovered from context; no target RGB or test LiDAR enters reconstruction.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian splatting [18]. Vehicles are represented as rigid assets with fixed local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The query layout assigns each latent code to a spatial location and scene component. Completion fills the unknown codes within this layout, and decoding preserves their component identities and motion bindings.

3.2. Visual recovery and canonical aggregation

The Driving Log Encoder extracts patch features from each image and exchanges pooled context across cameras and timestamps [9, 38]. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments, linking image features to 3D surfaces and their owning components. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations.

Spatial supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout retains canonical coordinates to identify component surfaces and first-input world coordinates $\mathbf{q}_i^w$, including human skinning, to locate them relative to the background and other actors.

Motion-aware evidence aggregation. To aggregate evidence for the same surface across time, we place each canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection gives query depth $z_{i,t,v}$ and samples image features $\boldsymbol{\phi}_{i,t,v}$, depth $d_{i,t,v}$, and confidence $c_{i,t,v}$. We weight observations by confidence and normalized depth residual $r = (d - z)/(\eta_{\text{abs}} + \eta_{\text{rel}} d)$. The indicator $a_{i,t,v}$ selects visible, depth-consistent observations of the same instance:

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v}[\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The masks m_i, f_i, and u_i distinguish observed surfaces, ray-labeled free space, and unknown content, with $m_i + f_i + u_i = 1$. Confidence is the maximum over valid observations; two reliable depth-ray votes establish a free-space label. Our main protocol uses center-ray voting (C); the footprint alternative (F) tests the projected cell extent. Sky features are aggregated by viewing direction using camera rotation and sky semantics.

3.3. Joint latent completion

Latent encoding. Type-specific encoders map the 129-dimensional aggregate and its spatial support, $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$, through a $135 \to 128 \to 128$ MLP with LayerNorm. All types use $D = 128$ codes, allowing joint completion while retaining query type, ownership, and frame. Partial and complete branches share this map; masks and confidence condition completion separately. Unobserved codes are zeroed, and observed codes are denoted $\mathbf{z}^{\text{obs}}$.

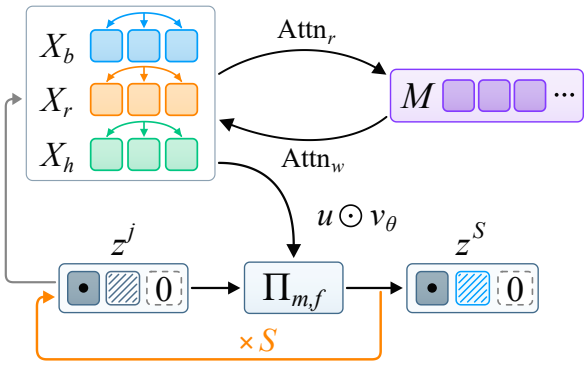

Figure 3. **Joint latent completion.** CCM reads local features and writes back shared context; projected flow updates only unknown codes.

Cross-component memory. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling (Fig. 3). We parameterize its completion velocity with a DiT-style transformer [30] trained by flow matching [22]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. Cross-component memory (CCM) then exchanges context across groups through learnable memory tokens, following induced set attention [17, 20]:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates local tokens; attention arguments are queries, keys, and values. Memory first reads all local groups, then supplies the updated scene context to each group's tokens. A residual feed-forward update follows $\mathbf{U}$. Flow time modulates local normalization, and learned tokens plus log summaries initialize memory.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1 - \tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \quad (9)$$

Prior learning uses samples with valid targets at every unknown query; s_i selects loss terms within these targets (Sec. 3.5).

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \quad (10)$$

After every step, $\Pi_{m,f}$ restores observed codes and clears free-space codes. The seed fixes a completion, decoded once for reuse across states.

Type-specific Gaussian decoding. Background, rigid, and human Gaussian decoders independently map each query to $K = 4$ child Gaussians whose centers lie in its support cell. Human children inherit the query's skinning weights; rigid children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}.

Because decoding is query-wise, freezing the codec and retaining observed codes preserves observed Gaussian parameters. Free-ray losses constrain the visibility and alpha-compositing effects of added Gaussians, measured in Sec. 4.4.

3.4. State-driven scene composition

At time step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. Rigid transforms update canonical Gaussian centers and covariances. Human joint poses deform them through the stored skinning weights, so completed surfaces move with the corresponding body parts. The renderer composites the background, active assets, and valid sky:

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \quad (11)$$

The background and canonical assets remain fixed; new camera and actor states reuse the same completed content.

3.5. Training procedure

We train recovery and the codec before learning the completion prior and adapting the GCM through rendering. Complete targets use the partial input's query layout $\mathcal{Q}$, including its cell scales, component ownership, and human bindings. Complete meshes supply surface samples within each cell. Multiview rendering and the frozen visual encoder produce $\mathbf{h}_i^{\text{full}}$, which the shared type-specific encoder maps to $\mathbf{z}_{1,i} = E_{k_i}([\mathbf{h}_i^{\text{full}}; \mathbf{q}_i; \boldsymbol{\delta}_i])$. Surface-free cells receive zero features and opacity supervision.

After codec fitting, recovery and the codec remain frozen. Stage 3 learns from complete synthetic targets at every unknown query. Stage 4 updates only the GCM, mixing complete synthetic samples and the 740 real training logs at 1:3. The four stages use 287,360 updates in total; Appendix B gives target construction, network sizes, and training settings.

4. Experiments

Waymo experiments test reconstruction accuracy and off-trajectory rendering quality. Paired hidden-surface tests and actor edits examine joint completion and reuse of fixed assets. Main results use center-ray rule C; the supplementary material compares footprint rule F and reports zero-shot transfer to nuScenes.

4.1. Experimental setup

Data and supervision. We divide Waymo's 798 training sequences into 740 training and 58 development sequences, evaluating on 202 validation sequences. Synthetic training uses 10,000 scenes, 80 maps, 10,000 vehicles, and 2,500 human identities. Source maps, vehicle models, and human identities are split before creating appearance/motion variants; real logs are separated by road segment, capture, and identifiable actors.

Baselines. STORM, DGGT, and StreetForward [5, 49, 52] are retrained from random initialization on the same real logs, synthetic scenes, and complete assets. Each uses its native supervision; Ours also uses cell-empty and template-binding targets. A shared state adapter assigns ownership and canonicalizes predicted Gaussians for rigid and skeletal placement. Methods share context RGB, calibration, ego poses, vehicle boxes/root tracks, and target human joints; shape and binding come from context. Test-time weights are fixed. DGGT denotes its Gaussian branch (DGGT-G).

Sampling and metrics. Each test sequence supplies 200 frames, forming 2,020 disjoint 20-frame clips at 10 Hz. Three front cameras provide context at $\{0, 5, 10, 15\}$ [49]; the remaining 12 interpolation and four extrapolation frames give 96,960 targets. Extrapolation renders supplied states. We evaluate 240×160 images [16] using full/dynamic PSNR and SSIM, and depth AbsRel/δ_1 at valid LiDAR points. Depth is aligned per frame for evaluation. Frames, clips, then sequences are averaged with equal sequence weight.

Figure 4. **Conditional dynamic reconstruction.** Rows show five examples; columns are ground truth, DGGT, STORM, StreetForward, and Ours.

4.2. Dynamic scene reconstruction

DriveCompose leads all six Waymo metrics (Table 1), exceeding DGGT by 1.28 and 1.68 dB in full-image and dynamic-region PSNR. With context-predicted instead of supplied actor states, dynamic-region PSNR drops to 23.83 dB but still exceeds DGGT by 1.14 dB (Appendix F). Figure 4 shows clearer actor silhouettes and street structure.

Table 1. **Conditional reconstruction on Waymo.** Interpolation and extrapolation are aggregated at 3:1.

	Full image		Dynamic Region		Scale-aligned depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	25.79	0.782	22.38	0.617	0.194	0.733
DGGT	27.35	0.875	23.49	0.643	0.238	0.702
StreetForward	25.88	0.819	21.34	0.604	0.216	0.721
Ours	**28.63**	**0.903**	**25.17**	**0.867**	**0.093**	**0.881**

4.3. Off-trajectory rendering

We shift cameras laterally by ±1, ±2, and ±4 m while fixing orientation, intrinsics, and timestamps [8, 55]. We compare Gaussian branches and full generative pipelines. Each direction contains 121,200 images and 6,060 20-frame videos at 10 Hz, referenced to recordings from the same clips, cameras, and timestamps. Left/right FID and FVD are computed separately, then averaged (Table 2).

Full generative controls include DGGT-Full and STORM+Difix [43]. Ours leads all six entries (Table 2) and has the lowest MAE/LPIPS on a fixed reference region (Appendix H). Figure 5 shows shifted renders.

4.4. State-driven asset reuse

Asset edits. We test fixed-asset reuse on twenty real scenes, each containing one vehicle and one human. A 1 m camera shift, 30° vehicle rotation, and human pose change reuse fixed completed assets. Nine training/generation-seed combinations yield 360 vehicle and human edits; camera shifts are scored separately. Structural integrity passes

Figure 5. **Off-trajectory rendering with a 2 m leftward shift.** Columns show GT, STORM, DGGT, StreetForward, and Ours. GT is the original-trajectory image.

Table 2. **Off-trajectory distribution quality.** FID/FVD are computed separately for left and right shifts, then averaged.

Method	Shift ±1 m		Shift ±2 m		Shift ±4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	65.27	756.43	87.63	1074.81	136.41	1762.37
DGGT-G	70.13	914.73	90.47	1135.81	127.69	1593.27
StreetForward	84.67	883.29	103.79	1103.67	125.31	1589.43
DGGT-Full	36.47	592.63	46.81	743.17	69.23	997.41
STORM+Difix	33.19	578.43	42.87	723.61	65.31	945.27
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

334/360 (92.78%); 298/360 (82.78%) jointly pass structure, new-surface texture, identity, and geometry, compared with 80.28% for Asset Harvester [2], which leads identity preservation by 0.83 points. Across six edit levels, joint acceptance is 74.54%, 4.91 points above Asset Harvester (Appendix I).

Observation preservation. Compared with the observed-only scene at fixed cameras and actor states, Ours preserves observed Gaussian parameters exactly, while input-view PSNR decreases by 0.03 dB and depth MAE increases by 2 mm. Previously observed human texture has canonical linear-RGB MAE 0.014.

Efficiency. On an RTX 4090 (FP16, batch 1, rule C), construction takes 2.66 ± 0.11 s, single-camera composition/rendering 7.8 ± 0.4 ms at 240×160, and peak memory 14.2 GB. Reusing the constructed assets amortizes the total cost over 1,000 outputs to 10.46 ms each. At 480×320 input and output, Ours exceeds every external method by at least 1.49 dB full-image PSNR and renders one image in 13.71 ms, versus 161.87 ms for DGGT-Full (Appendix K).

4.5. Ablation studies

Waymo ablations use one 29-timestamp window per test sequence, three cameras, and seven shared target timestamps.

Completion components. The w/o GCM control disables non-observed, non-sky Gaussians. The w/o CCM control uses isolated per-component memories with the same parameters and global log context, retraining Stages 3–4. Variants share recovery, codec, states, data, and optimization budgets. Training samples T uniformly; inference uses 24 Euler steps. The learned variants average three training and three generation seeds, retaining every output; w/o GCM is deterministic. At $T = 8$, exchanging component context adds 0.627/0.934 dB full-image/dynamic PSNR (Table 3). Figure 6 shows cleaner traffic-signal contours and sharper foliage detail with the full model. Pedestrian silhouettes and vehicle windows have fewer distortions and ghosting artifacts. On 100 synthetic scenes with paired hidden-surface references, shared context exceeds task-adapted object-coupling and scene-latent completion by 0.67 and 0.94 dB hidden-region PSNR, and its gain rises from 0.03–0.05 dB on well-observed surfaces to 1.02–1.53 dB on severely missing ones. With matched queries, dense attention is 0.05 dB higher, while CCM completes 1.93× faster with 42.71% lower peak memory (Appendix H).

Table 3. **Component ablations on Waymo.** $T = 8$ timestamps (24 images), with shared recovery, codec, and target views. Full metrics and SDs are in Appendix G.

Variant	Full image			Dynamic
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑
w/o GCM	27.137	0.867	0.183	23.431
w/o CCM	28.286	0.893	0.147	24.683
Ours	**28.913**	**0.911**	**0.129**	**25.617**

Input-timestamp sensitivity. Fixed checkpoints use nested $T = 2, 4, 8$ inputs (6/12/24 images) over the same time span and target views. At each input count, evidence and queries are reconstructed from the supplied inputs. CCM adds 1.104, 0.738, and 0.627 dB PSNR and reduces LPIPS by 0.041, 0.028, and 0.018 (Table 4), with the largest benefit at the sparsest input setting.

Table 4. **Input-timestamp sensitivity.** Fixed checkpoints, time span, and target views; T counts timestamps, each with three images.

T	Variant	PSNR ↑	SSIM ↑	LPIPS ↓
2	w/o CCM	25.237	0.817	0.254
2	Ours	**26.341**	**0.851**	**0.213**
4	w/o CCM	27.126	0.869	0.187
4	Ours	**27.864**	**0.891**	**0.159**
8	w/o CCM	28.286	0.893	0.147
8	Ours	**28.913**	**0.911**	**0.129**

(a) Ours (b) w/o CCM (c) w/o GCM

Figure 6. **Completion component ablations.** Rows show two scenes; columns are Ours, w/o CCM, and w/o GCM. Colored boxes mark the regions enlarged below each view.

5. Discussion

DriveCompose reconstructs finite regions under fixed illumination, using rigid vehicles and humans bound to a fixed skeleton and clothing volume. Depth, identity, and canonical-pose accuracy determine the observed anchors inherited by completion. Residual reconstruction artifacts occur near pedestrians and the parked pickup in Fig. 4. Editing failures concentrate on large elbow bends, leg self-occlusion, and loose-clothing boundaries. Relighting and clothing-geometry changes call for richer appearance and deformation models.

6. Conclusion

DriveCompose jointly completes world-space background and canonical actor latents, preserving component identity and motion bindings for state-driven reuse. On paired synthetic tests, shared scene context improves hidden geometry and appearance over task-adapted scene-latent and object-coupling controls. Fixed Gaussian assets support camera shifts, vehicle rotation, and human pose edits. Waymo reconstruction, paired surface metrics, and multidimensional real-edit ratings establish completion quality and state-driven reuse.

LLM Usage Statement

OpenAI Codex was used solely for language polishing of the manuscript, including grammar correction and improvements to clarity and readability. The authors take full responsibility for the final manuscript.

References

[1] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuScenes: A Multimodal Dataset for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11621–11631, 2020. 1

[2] Tianshi Cao, Jiawei Ren, Yuxuan Zhang, Jaewoo Seo, Jiahui Huang, Shikhar Solanki, Haotian Zhang, Mingfei Guo, Haithem Turki, Muxingzi Li, Yue Zhu, Sipeng Zhang, Zan Gojcic, Sanja Fidler, and Kangxue Yin. Asset harvester: Extracting 3d assets from autonomous driving logs for simulation. *arXiv preprint arXiv:2604.18468*, 2026. 7

[3] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3D Gaussian Splats from Image Pairs for Scalable Generalizable 3D Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19457–19467, 2024. 2

[4] Shijie Chen and Peixi Peng. Freegen: Feed-forward reconstruction–generation co-training for free-viewpoint driving scene synthesis. In *Computer Vision – ECCV 2026*, pages 682–699. Springer, 2026. 2

[5] Xiaoxue Chen, Ziyi Xiong, Yuantao Chen, Gen Li, Nan Wang, Hongcheng Luo, Long Chen, Haiyang Sun, Bing Wang, Guang Chen, Hongyang Li, Ya-Qin Zhang, Hangjun Ye, and Hao Zhao. DGGT: Feedforward 4D Reconstruction of Dynamic Driving Scenes using Unposed Images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1265–1276, 2026. 2, 5

[6] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. In *European Conference on Computer Vision (ECCV)*, 2024. 2

[7] Ziyu Chen, Jiawei Yang, Jiahui Huang, Riccardo de Lutio, Janick Martinez Esturo, Boris Ivanovic, Or Litany, Zan Gojcic, Sanja Fidler, Marco Pavone, Li Song, and Yue Wang. OmniRe: Omni Urban Scene Reconstruction. In *International Conference on Learning Representations*, pages 85508–85527, 2025. 2

[8] Cheng Chi, Xianqi Wang, Hongcheng Luo, Mingfei Tu, Gangwei Xu, Zehan Zhang, Bing Wang, Guang Chen, Hangjun Ye, Sida Peng, Xin Yang, and Haiyang Sun. PointForward: Feedforward Driving Reconstruction through Point-Aligned Representations. *arXiv preprint arXiv:2605.11594*, 2026. 2, 6

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*, 2021. 3

[10] Ruiyuan Gao, Kai Chen, Enze Xie, Lanqing HONG, Zhenguo Li, Dit-Yan Yeung, and Qiang Xu. MagicDrive: Street View Generation with Diverse 3D Geometry Control. In *International Conference on Learning Representations*, pages 22841–22860, 2024. 2

[11] Ruiqi Gao, Aleksander Hołyński, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul Srinivasan, Jonathan T. Barron, and Ben Poole. CAT3D: Create Anything in 3D with Multi-View Diffusion Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 75468–75494, 2024. 2

[12] Shenyuan Gao, Jiazhi Yang, Li Chen, Kashyap Chitta, Yihang Qiu, Andreas Geiger, Jun Zhang, and Hongyang Li. Vista: A Generalizable Driving World Model with High Fidelity and Versatile Controllability. In *Advances in Neural Information Processing Systems*, pages 91560–91596, 2024. 2

[13] Georg Hess, Carl Lindström, Maryam Fatemi, Christoffer Petersson, and Lennart Svensson. SplatAD: Real-Time Lidar and Camera Rendering with 3D Gaussian Splatting for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11982–11992, 2025. 2

[14] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large Reconstruction Model for Single Image to 3D. In *International Conference on Learning Representations*, pages 50678–50702, 2024. 2

[15] Shoukang Hu, Tao Hu, and Ziwei Liu. GauHuman: Articulated Gaussian Splatting from Monocular Human Videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20418–20431, 2024. 2

[16] Jiahui Huang, Jiawei Ren, Michal Tyszkiewicz, Bjoern Haefner, Michael Shelley, Xin Kang, Seung Wook Kim, Ning Xu, Qi Wu, Janick Martinez Esturo, Shengyu Huang, Nick Schneider, Laura Leal-Taixé, Zan Gojcic, and Sanja Fidler. Instant NuRec: Feed-Forward 3D Gaussian Reconstruction for Driving Scene Simulation. *arXiv preprint arXiv:2607.14203*, 2026. 2, 5

[17] Andrew Jaegle, Felix Gimeno, Andy Brock, Oriol Vinyals, Andrew Zisserman, and Joao Carreira. Perceiver: General Perception with Iterative Attention. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 4651–4664, 2021. 4

[18] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4), 2023. 2, 3

[19] Muhammed Kocabas, Jen-Hao Rick Chang, James Gabriel, Oncel Tuzel, and Anurag Ranjan. HUGS: Human Gaussian Splats. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 505–515, 2024. 2

[20] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosiorek, Seungjin Choi, and Yee Whye Teh. Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 3744–3753, 2019. 4

[21] Ziwei Liao, Mohamed Sayed, Steven L. Waslander, Sara Vicente, Daniyar Turmukhambetov, and Michael Firman. Complete Gaussian Splats from a Single Image with Denoising Diffusion Models. *arXiv preprint arXiv:2508.21542*, 2025. 2

[22] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow Matching for Generative Modeling. In *International Conference on Learning Representations*, 2023. 2, 4

[23] Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot One Image to 3D Object. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9298–9309, 2023. 2

[24] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A Skinned Multi-Person Linear Model. *ACM Trans. Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, 2015. 2

[25] Yichong Lu, Yichi Cai, Shangzhan Zhang, Hongyu Zhou, Haoji Hu, Huimin Yu, Andreas Geiger, and Yiyi Liao. UrbanCAD: Towards Highly Controllable and Photorealistic 3D Vehicles for Urban Scene Simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27519–27530, 2025. 2

[26] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *European Conference on Computer Vision (ECCV)*, 2020. 2

[27] Julian Ost, Fahim Mannan, Nils Thuerey, Julian Knodt, and Felix Heide. Neural Scene Graphs for Dynamic Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2856–2865, 2021. 2

[28] Jinhyung Park, Navyata Sanghvi, Erica Weng, Shawn Hunt, Shinya Tanaka, Hironobu Fujiyoshi, and Kris Kitani. Grounded Latents for Entity-Centric 4D Scene Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21420–21430, 2026. 2

[29] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive Body Capture: 3D Hands, Face, and Body From a Single Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10975–10985, 2019. 2

[30] William Peebles and Saining Xie. Scalable Diffusion Models with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023. 4

[31] Zhiyin Qian, Shaofei Wang, Marko Mihajlovic, Andreas Geiger, and Siyu Tang. 3DGS-Avatar: Animatable Avatars via Deformable 3D Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5020–5030, 2024. 2

[32] Lingteng Qiu, Xiaodong Gu, Peihao Li, Qi Zuo, Weichao Shen, Junfei Zhang, Kejie Qiu, Weihao Yuan, Guanying Chen, Zilong Dong, and Liefeng Bo. LHM: Large Animatable Human Reconstruction Model for Single Image to 3D in Seconds. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14184–14194, 2025. 2

[33] Jiawei Ren, Michal Jan Tyszkiewicz, Jiahui Huang, and Zan Gojcic. TokenGS: Decoupling 3D Gaussian Prediction from Pixels with Learnable Tokens. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15365–15375, 2026. 2

[34] Yichun Shi, Peng Wang, Jianglong Ye, Long Mai, Kejie Li, and Xiao Yang. MVDream: Multi-view Diffusion for 3D Generation. In *International Conference on Learning Representations*, pages 39838–39859, 2024. 2

[35] Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, Vijay Vasudevan, Wei Han, Jiquan Ngiam, Hang Zhao, Aleksei Timofeev, Scott Ettinger, Maxim Krivokon, Amy Gao, Aditya Joshi, Sheng Zhao, Shuyang Cheng, Yu Zhang, Jonathon Shlens, Zhifeng Chen, and Dragomir Anguelov. Scalability in Perception for Autonomous Driving: Waymo Open Dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2446–2454, 2020. 1

[36] Qijian Tian, Xin Tan, Yuan Xie, and Lizhuang Ma. DrivingForward: Feed-forward 3D Gaussian Splatting for Driving Scene Reconstruction from Flexible Surround-view Input. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 7374–7382, 2025. 2

[37] Adam Tonderski, Carl Lindström, Georg Hess, William Ljungbergh, Lennart Svensson, and Christoffer Petersson. NeuRAD: Neural Rendering for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14895–14904, 2024. 2

[38] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is All you Need. In *Advances in Neural Information Processing Systems*, 2017. 3

[39] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual Geometry Grounded Transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5294–5306, 2025. 2

[40] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3D Vision Made Easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20697–20709, 2024. 2

[41] Xiaofeng Wang, Zheng Zhu, Guan Huang, Xinze Chen, Jiagang Zhu, and Jiwen Lu. DriveDreamer: Towards Real-world-driven World Models for Autonomous Driving. In *European Conference on Computer Vision (ECCV)*, 2024. 2

[42] Yifan Wang, Francesco Pittaluga, Zaid Tasneem, Chenyu You, Manmohan Chandraker, and Ziyu Jiang. HorizonForge: Driving Scene Editing with Any Trajectories and Any Vehicles. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[43] Jay Zhangjie Wu, Yuxuan Zhang, Haithem Turki, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic, and Huan Ling. DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26024–26035, 2025. 2, 6

[44] Qirui Wu, Yawar Siddiqui, Duncan Frost, Samir Aroudj, Armen Avetisyan, Richard Newcombe, Angel X. Chang, Jakob Engel, and Henry Howard-Jenkins. JRM: Joint Reconstruction Model for Multiple Objects without Alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 307–316, 2026. 2

[45] Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. Structured 3D Latents for Scalable and Versatile 3D Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21469–21480, 2025. 2

[46] Jianfeng Xiang, Xiaoxue Chen, Sicheng Xu, Ruicheng Wang, Zelong Lv, Yu Deng, Hongyuan Zhu, Yue Dong, Hao Zhao, Nicholas Jing Yuan, and Jiaolong Yang. Native and Compact Structured Latents for 3D Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14419–14429, 2026. 2

[47] Yunzhi Yan, Haotong Lin, Chenxu Zhou, Weijie Wang, Haiyang Sun, Kun Zhan, Xianpeng Lang, Xiaowei Zhou, and Sida Peng. Street Gaussians: Modeling Dynamic Urban Scenes with Gaussian Splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 2

[48] Yunzhi Yan, Zhen Xu, Haotong Lin, Haian Jin, Haoyu Guo, Yida Wang, Kun Zhan, Xianpeng Lang, Hujun Bao, Xiaowei Zhou, and Sida Peng. StreetCrafter: Street View Synthesis with Controllable Video Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 822–832, 2025. 2

[49] Jiawei Yang, Jiahui Huang, Boris Ivanovic, Yuxiao Chen, Yan Wang, Boyi Li, Yurong You, Apoorva Sharma, Maximilian Igl, Peter Karkus, Danfei Xu, Yue Wang, and Marco Pavone. STORM: Spatio-TempOral Reconstruction Model For Large-Scale Outdoor Scenes. In *International Conference on Learning Representations*, pages 50446–50465, 2025. 2, 5

[50] Ze Yang, Yun Chen, Jingkang Wang, Sivabalan Manivasagam, Wei-Chiu Ma, Anqi Joyce Yang, and Raquel Urtasun. UniSim: A Neural Closed-Loop Sensor Simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1389–1399, 2023. 1, 2

[51] Ze Yang, Jingkang Wang, Haowei Zhang, Sivabalan Manivasagam, Yun Chen, and Raquel Urtasun. GenAssets: Generating in-the-wild 3D Assets in Latent Space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22392–22403, 2025. 2

[52] Zhongrui Yu, Zhao Wang, Yijia Xie, Yida Wang, Xueyang Zhang, Yifei Zhan, and Kun Zhan. StreetForward: Perceiving Dynamic Street with Feedforward Causal Attention. *arXiv preprint arXiv:2603.19552*, 2026. 2, 5

[53] Hongyu Zhou, Longzhong Lin, Jiabao Wang, Yichong Lu, Dongfeng Bai, Bingbing Liu, Yue Wang, Andreas Geiger, and Yiyi Liao. HUGSIM: A Real-Time, Photo-Realistic and Closed-Loop Simulator for Autonomous Driving. *arXiv preprint arXiv:2412.01718*, 2024. 1, 2

[54] Xiaoyu Zhou, Zhiwei Lin, Xiaojun Shan, Yongtao Wang, Deqing Sun, and Ming-Hsuan Yang. DrivingGaussian: Composite Gaussian Splatting for Surrounding Dynamic Autonomous Driving Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21634–21643, 2024. 2

[55] Ziyue Zhu, Zhanqian Wu, Zhenxin Zhu, Lijun Zhou, Haiyang Sun, Bing Wan, Kun Ma, Guang Chen, Hangjun Ye, Jin Xie, and Jian Yang. WorldSplat: Gaussian-Centric Feed-Forward 4D Scene Generation for Autonomous Driving. *arXiv preprint arXiv:2509.23402*, 2025. 2, 6

[56] Yiyu Zhuang, Jiaxi Lv, Hao Wen, Qing Shuai, Ailing Zeng, Hao Zhu, Shifeng Chen, Yujiu Yang, Xun Cao, and Wei Liu. IDOL: Instant Photorealistic 3D Human Creation from a Single Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26308–26319, 2025. 2